

IPR UNIT-5

1. Discuss about E-Commerce its pros and cons

A. **E-Commerce** (Electronic Commerce) refers to the buying and selling of goods and services over the internet. It includes:

- Online retail (e.g., Amazon, Flipkart)
- Digital services (e.g., Netflix, online courses)
- B2B platforms (e.g., Alibaba)
- Mobile commerce and electronic payments

Pros of E-Commerce

- **Convenience:** Customers can shop 24/7 from anywhere.
- **Wider reach:** Sellers can access national and global markets.
- **Lower costs:** Reduced need for physical stores and staff.
- **Personalization:** AI-driven recommendations and targeted marketing.
- **Inventory efficiency:** Real-time tracking and automated restocking.

Cons of E-Commerce

- **Security risks:** Vulnerable to hacking, fraud, and data breaches.
- **Lack of physical inspection:** Customers can't touch or test products.
- **Logistics challenges:** Delays, damaged goods, and return complexities.
- **Digital divide:** Limited access for rural or low-income users.
- **Trust issues:** Fake websites, poor customer service, and misleading listings.

2. Write in detail about the benefits of data security

A. 1. Preserves Confidentiality

Data security ensures that sensitive information—like personal details, financial records, or business secrets—is accessible only to authorized users. → This protects individuals from identity theft and organizations from industrial espionage.

2. Maintains Data Integrity

Security mechanisms prevent unauthorized alterations, deletions, or corruption of data. → This ensures that data remains accurate, reliable, and trustworthy for decision-making.

3. Ensures Availability

By protecting systems from attacks (e.g., denial-of-service), data security helps maintain uninterrupted access to resources. → This supports business continuity and operational efficiency.

4. Builds Trust and Reputation

Organizations that safeguard user data earn customer trust and loyalty. → This enhances brand reputation and reduces churn.

5. Supports Legal Compliance

Data protection laws (like the IT Act, GDPR, HIPAA) require organizations to secure personal and sensitive data. → Compliance avoids penalties and legal liabilities.

6. Prevents Financial Loss

Security breaches can lead to theft, fraud, and costly recovery efforts. → Strong data

security minimizes risk and protects assets.

3. Explain about data Authentication and Confidentiality

A. 1. Data Authentication

Definition: Authentication is the process of verifying the identity of a user, device, or system before granting access to data or services.

Purpose: To ensure that the entity requesting access is genuine and not an imposter.

Common Methods:

- **Passwords and PINs**
- **Biometric verification** (fingerprint, iris scan)
- **Two-factor authentication (2FA)** (e.g., OTP + password)
- **Digital signatures** (for verifying sender identity in electronic documents)

Example: When logging into a bank app, entering your password and receiving an OTP ensures that only you can access your account.

2. Data Confidentiality

Definition: Confidentiality ensures that data is accessible only to authorized individuals and is protected from unauthorized disclosure.

Purpose: To prevent sensitive information from being exposed, leaked, or intercepted.

Techniques Used:

- **Encryption** (e.g., AES, RSA)
- **Access control lists (ACLs)**
- **Secure communication protocols** (e.g., HTTPS, SSL/TLS)
- **Data masking and anonymization**

Example: When you send a message over WhatsApp, end-to-end encryption ensures that only the sender and recipient can read it.

4. Illustrate cyber-crime and its various forms with examples

A. Cyber-crime refers to illegal activities carried out using computers, networks, or digital devices. It targets individuals, organizations, or governments and often involves data theft, financial fraud, or disruption of services.

Major Forms of Cyber-Crime with Examples

a) Hacking

Unauthorized access to computer systems or networks. → Example: A hacker breaks into a bank's server to steal customer data.

b) Phishing

Deceptive emails or websites trick users into revealing personal information. → Example: A fake email from "your bank" asks for your login credentials.

c) Identity Theft

Stealing personal data to impersonate someone. → Example: Using stolen Aadhaar details to open fraudulent accounts.

d) Cyberstalking

Online harassment or surveillance of individuals. → Example: Repeated threatening

messages sent via social media.

e) Ransomware Attacks

Malware encrypts files and demands payment for decryption. → Example: A hospital's patient records are locked until ransom is paid.

f) Online Fraud

Fake websites or scams that trick users into financial loss. → Example: A fake e-commerce site collects payments but never delivers goods.

g) Child Pornography and Exploitation

Distribution or possession of illegal content involving minors. → Example: Sharing explicit content through hidden forums or encrypted apps.

h) Cyber Terrorism

Attacks on critical infrastructure to cause panic or disruption. → Example: Disabling power grids or government networks during conflict.

5. Discuss the liability of the internet service provider

A. An **Internet Service Provider (ISP)** is a company or entity that offers internet access and related services to users. Examples include BSNL, Airtel, Jio, and ACT. ISPs act as **intermediaries**—they transmit, host, or store data created by users but do not originate it.

Legal Framework – IT Act, 2000 (Section 79)

Under Section 79 of the **Information Technology Act, 2000**, ISPs are granted **safe harbor protection**, meaning:

- They are **not liable** for third-party content if:
 - They do not initiate the transmission.
 - They do not select the receiver.
 - They do not modify the content.

This protects ISPs from being punished for illegal content posted by users—unless they are complicit.

When ISPs Become Liable

ISPs lose protection and become liable if:

- They **knowingly host or transmit** unlawful content.
- They **fail to act** after receiving actual knowledge or a government/court order to remove illegal content.
- They **ignore due diligence obligations**, such as content moderation or grievance redressal.

Example: If an ISP continues to host pirated movies after being notified, it may face civil or criminal liability.

Duties of ISPs

- Follow **due diligence guidelines** under IT Rules.
- Respond to **takedown notices** from courts or authorities.
- Maintain **records and logs** for investigation.
- Cooperate with **law enforcement agencies**.

6. Illustrate about digital signature, its creation and benefits

A. A **digital signature** is a cryptographic technique used to validate the authenticity and integrity of a digital message or document. It is the electronic equivalent of a handwritten signature or a stamped seal, but far more secure.

It ensures:

- The message was created by a known sender (authentication)
- The message was not altered in transit (integrity)
- The sender cannot deny sending it (non-repudiation)

Creation of Digital Signature

Let's walk through the process:

1. **Hashing the message:** The original document is passed through a hash function to generate a fixed-length digest (e.g., SHA-256).
2. **Encrypting the hash:** The sender encrypts the hash using their **private key**. This encrypted hash is the digital signature.
3. **Sending the message and signature:** The original message and the digital signature are sent together.
4. **Verification by receiver:**
 - The receiver decrypts the signature using the sender's **public key** to retrieve the original hash.
 - They also hash the received message independently.
 - If both hashes match, the signature is valid.

This process uses **asymmetric cryptography** (public-private key pair) and ensures that only the sender could have signed the message.

Benefits of Digital Signatures

- **Authentication:** Confirms the identity of the sender.
- **Integrity:** Detects any tampering with the message.
- **Non-repudiation:** Prevents the sender from denying the transaction.
- **Legal validity:** Recognized under the **Information Technology Act, 2000** in India.
- **Efficiency:** Enables secure, paperless transactions in e-governance, banking, and e-commerce.

7. Discuss Data security and its benefits.

A. **Data security** refers to the protection of digital information from unauthorized access, corruption, theft, or loss. It ensures that data remains confidential, accurate, and available to authorized users only.

2. Key Benefits of Data Security

Let's go one by one:

a) Confidentiality Only authorized users can access sensitive data. → Prevents identity theft, espionage, and data leaks.

b) Integrity Ensures data is not altered or tampered with. → Supports reliable decision-making and system trust.

c) Availability Protects systems from disruptions (e.g., DDoS attacks). → Ensures continuous access to data and services.

d) Legal Compliance Meets requirements under laws like the IT Act, GDPR, HIPAA.

→ Avoids penalties and builds legal credibility.

e) Trust and Reputation Organizations that secure data earn user confidence. → Enhances brand value and customer loyalty.

f) Financial Protection Prevents losses due to fraud, ransomware, or recovery costs. → Supports business continuity and risk management.

8. Examine Privacy and the importance of data privacy

A. Privacy refers to an individual's right to control access to their personal information, communications, and choices. It includes:

- Physical privacy (freedom from surveillance)
- Informational privacy (control over personal data)
- Decisional privacy (freedom to make personal choices)

Data privacy is a subset of privacy focused on how personal data is collected, stored, processed, and shared. It ensures that:

- Data is used lawfully and transparently
- Individuals retain control over their information
- Sensitive data is protected from misuse

Importance of Data Privacy

- **Protects Personal Information:** Data privacy safeguards individuals' sensitive information, preventing identity theft, fraud, and unauthorized use of their data.
- **Maintains Trust:** Organizations that prioritize data privacy build consumer trust and enhance their reputation by demonstrating responsibility in handling personal data.
- **Legal Compliance:** Data privacy helps businesses comply with laws and regulations such as GDPR and CCPA, avoiding fines and legal penalties.
- **Preserves Individual Autonomy:** It empowers individuals to control what personal data they share and how it is used, ensuring their privacy rights are respected.
- **Prevents Reputational Damage:** Breaches or misuse of personal data can lead to severe damage to an organization's image, making privacy a critical component of risk management.

Data privacy is essential to protect individuals' identities and sensitive information, maintain trust in digital interactions, comply with legal requirements, and uphold ethical standards in data handling.

9. Write in detail about cyber-crime laws, prevention and punishment in India

A. 1. Legal Framework Governing Cybercrime in India

India primarily addresses cybercrime through the **Information Technology Act, 2000 (IT Act)**, supplemented by relevant provisions of the **Indian Penal Code (IPC)**.

- The IT Act provides legal recognition to electronic records and digital signatures, and criminalizes various cyber offences.
- The **2008 amendment** expanded its scope to include cyber terrorism, identity theft, and data breaches.
- The **Digital Personal Data Protection Act, 2023 (DPDPA)** adds a layer of data

privacy enforcement, especially for personal data processing.

2. Major Cyber Offences under Indian Law

- **Hacking and unauthorized access:** Section 66 of the IT Act penalizes unauthorized access to computer systems and data manipulation.
- **Identity theft and impersonation:** Section 66C criminalizes fraudulent use of digital credentials like passwords or biometric data.
- **Cyberstalking and voyeurism:** Section 66E of the IT Act and Section 354D of the IPC address capturing or transmitting private images without consent.
- **Cyber terrorism:** Section 66F punishes acts that threaten national security via cyber means, including attacks on critical infrastructure.
- **Obscene content and pornography:** Section 67 penalizes publishing or transmitting obscene material electronically, while Section 67B specifically targets child sexual abuse material.
- **Data breach and confidentiality violations:** Section 72 penalizes service providers who disclose personal information without consent.

3. Preventive Measures

Governmental Initiatives:

- **CERT-In (Indian Computer Emergency Response Team)** monitors and responds to cybersecurity incidents and issues advisories.
- **Cyber Swachhta Kendra** promotes tools for malware detection and system cleanup.
- **National Cyber Crime Reporting Portal** enables online reporting of cyber offences.

Institutional Safeguards:

- Critical infrastructure entities are required to conduct regular cyber audits.
- Organizations must implement data protection policies aligned with the DPDPA.
- Law enforcement agencies are being trained in digital forensics and cyber investigation.

Individual Best Practices:

- Use of strong passwords, two-factor authentication, and secure browsing habits.
- Avoiding suspicious links, phishing emails, and unverified downloads.
- Prompt reporting of cyber incidents to local cybercrime cells or online portals.

4. Punishment and Enforcement Mechanisms

- **Adjudicating Officers** handle civil violations such as data breaches and impose fines.
- **Cybercrime Police Stations** and specialized units under the **CBI** investigate serious cyber offences.
- **Courts** may impose imprisonment, fines, or both, depending on the nature and gravity of the offence.
- **Juvenile offenders** are dealt with under the Juvenile Justice Act, focusing on reform rather than punishment.

5. Challenges and Emerging Trends

- **Jurisdictional complexity** in cross-border cybercrimes complicates investigation and prosecution.
- **Underreporting** remains a concern due to lack of awareness or fear of reputational damage.
- **Evolving threats** such as ransomware, deepfakes, and AI-driven frauds demand continuous legal and technical adaptation.

10. Summarize the protection of online and computer transactions in detail

A. Online and computer transactions in India are protected through a combination of legal safeguards, cybersecurity protocols, and user awareness initiatives. The IT Act, RBI guidelines, and CERT-In advisories form the backbone of this protection.

1. Legal and Regulatory Safeguards

- **Information Technology Act, 2000:** Provides legal recognition to electronic transactions and penalizes cyber fraud, hacking, and data theft.
- **Digital Personal Data Protection Act, 2023:** Mandates secure handling of personal data by organizations, with penalties for breaches.
- **Indian Penal Code (IPC):** Sections like 420 (cheating) and 468 (forgery) apply to online financial frauds.
- **RBI Guidelines:** Banks must implement multi-factor authentication, transaction alerts, and grievance redressal mechanisms for digital payments.

2. Institutional Protection Mechanisms

- **CERT-In (Indian Computer Emergency Response Team):** Monitors cyber threats and issues advisories to protect digital infrastructure.
- **Cybercrime Cells and Reporting Portals:** Citizens can report frauds via the National Cyber Crime Reporting Portal (cybercrime.gov.in).
- **Banking Ombudsman and Consumer Forums:** Provide redressal for unresolved complaints related to online transactions.

3. Technical and Security Measures

- **End-to-End Encryption:** Used in banking apps and payment gateways to secure data transmission.
- **Secure Socket Layer (SSL):** Ensures safe browsing and protects sensitive information during online transactions.
- **Two-Factor Authentication (2FA):** Adds an extra layer of security for login and payment verification.
- **Tokenization:** Replaces sensitive card details with encrypted tokens during transactions.

4. User-Level Best Practices

- **Use strong, unique passwords** and change them regularly.
- **Avoid public Wi-Fi** for financial transactions.
- **Verify URLs and app sources** before entering credentials.
- **Enable transaction alerts** via SMS/email to monitor activity.
- **Report suspicious activity immediately** to banks or cybercrime portals.

5. Common Threats and Prevention

- **Phishing and OTP scams:** Avoid clicking on unknown links or sharing OTPs.

- **Fake apps and websites:** Download only from trusted sources like Google Play or Apple App Store.
- **SIM swapping and identity theft:** Use biometric locks and monitor account changes.

11. Discuss about Certifying Authorities (CA), functions of controller and Certificate Authority (CA)

A. Certifying Authorities (CAs) are trusted organizations responsible for issuing and managing digital certificates that authenticate the identity of individuals, organizations, or devices in electronic transactions. In India, CAs operate under the supervision of the Controller of Certifying Authorities (CCA) established under the IT Act, 2000.

Functions of Certifying Authority (CA)

- **Issuance of Digital Certificates:** CAs verify the identity of applicants through documents, physical or electronic verification and issue digital signature certificates that bind public keys to the entity's identity.
- **Verification and Authentication:** They ensure that digital certificates are issued only to legitimate entities after careful identity validation.
- **Certificate Management:** Includes renewal, suspension, and revocation of certificates to maintain trust in the system.
- **Maintaining Repositories:** CAs maintain a repository of issued certificates and Certificate Revocation Lists (CRLs) accessible to users for verification of certificate validity.
- **Provisioning Sub-CAs:** CAs may create technical sub-CAs for infrastructure management but these are not independent legal entities.

Role and Functions of the Controller of Certifying Authorities (CCA)

- The CCA is the government authority that licenses and regulates CAs in India.
- Acts as the Root Certifying Authority of India (RCAI) to digitally sign and certify the public keys of licensed CAs, thereby establishing a trusted root of the digital certificate chain.
- The CCA ensures that CAs comply with the IT Act and its rules, and follows protocols regarding issuance, suspension, and revocation of digital certificates.
- Maintains policies, security standards, and the certification practice statements that CAs must adhere to for trusted digital signature issuance.
- Provides oversight to maintain the integrity and trustworthiness of the digital certification ecosystem in India.

Certifying Authorities are crucial for providing the digital credentials needed for secure online transactions by verifying identities and issuing digital certificates. The Controller of Certifying Authorities regulates and oversees all licensed CAs to ensure secure, reliable, and legally valid digital certification infrastructure in India.

12. Illustrate about the Information Technology Act, 2000 (IT Act) and its positive and negative aspects

A. Overview of the IT Act, 2000

- Enacted to provide legal recognition to electronic transactions and digital signatures.
- Facilitates e-commerce, e-governance, and cybercrime regulation.
- Applies to the whole of India and also to offences committed outside India involving Indian systems.
- Amended in **2008** to include cyber terrorism, identity theft, and data protection provisions.

Key Provisions

- **Legal recognition of electronic records and digital signatures** (Sections 3–10).
- **Regulation of Certifying Authorities** (Sections 17–34).
- **Offences and penalties** (Sections 43–78), including hacking, data theft, cyber terrorism, and publishing obscene content.
- **Role of Controller of Certifying Authorities (CCA)** in licensing and oversight.
- **Adjudication and appellate mechanisms** for cyber disputes.

Positive Aspects

1. **Legal Enablement of E-Commerce**
 - Recognizes contracts and signatures in digital form, boosting online business and banking.
2. **Cybercrime Deterrence**
 - Penalizes hacking, identity theft, and cyber terrorism, creating a legal deterrent.
3. **Digital Governance Support**
 - Facilitates secure e-governance services like e-filing, e-tendering, and digital certificates.
4. **Global Applicability**
 - Covers offences committed outside India if systems in India are affected.
5. **Flexibility and Amendability**
 - Has been updated (e.g., 2008 amendment) to address emerging threats like phishing and data breaches.

Negative Aspects

1. **Limited Scope in Early Versions**
 - Original Act lacked provisions for cyber terrorism, data protection, and privacy.
2. **Ambiguity in Definitions**
 - Terms like “obscenity” or “reasonable security practices” are vaguely defined, leading to interpretational issues.
3. **Enforcement Challenges**
 - Lack of trained cybercrime personnel and digital forensic infrastructure hampers effective enforcement.
4. **Jurisdictional Complexity**

- Cross-border cybercrimes pose legal and procedural hurdles.

5. Privacy Concerns

- Surveillance powers under Section 69 (interception and monitoring) raise civil liberty concerns.